K-Nearest Neighbors (KNN) — Notes

1. What is KNN?

KNN is a supervised ML algorithm mainly used for classification.

Main idea:

Similar data points are usually close to each other.

For a new point, KNN looks at its **K nearest points** and uses **majority voting** to decide its class.

---

2. How KNN Works

New point

↓

Calculate distance from training points

↓

Find K nearest points

↓

Majority vote

↓

Prediction

Example:

K = 5

3 neighbors → Class A

2 neighbors → Class B

Prediction → Class A

---

3. Distance

KNN commonly uses **Euclidean Distance**:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The smaller the distance, the **more similar** the points are.

---

4. Why Feature Scaling?

Suppose:

Age → 20–60

Salary → 20,000–1,00,000

Salary has much larger values, so it can dominate the distance.

Therefore, we usually scale features using:

```
from sklearn.preprocessing import StandardScaler


scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)
```

---

5. Choosing K

Small K:

K = 1

→ Sensitive to noise
→ Can **overfit**

Large K:

K = 100

→ Too generalized
→ Can **underfit**

Usually, we try different K values using **cross-validation** and choose the best one.

---

6. Important Limitations

- Slow prediction for **large datasets**
- Sensitive to **feature scaling**
- Can suffer from **curse of dimensionality**
- Sensitive to **outliers/noise**
- Can be biased with **imbalanced classes**

🧠 **Remember**

KNN = Find nearest neighbors → Majority vote → Prediction

Small K → Overfitting

Large K → Underfitting

Scaling → Very important